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CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

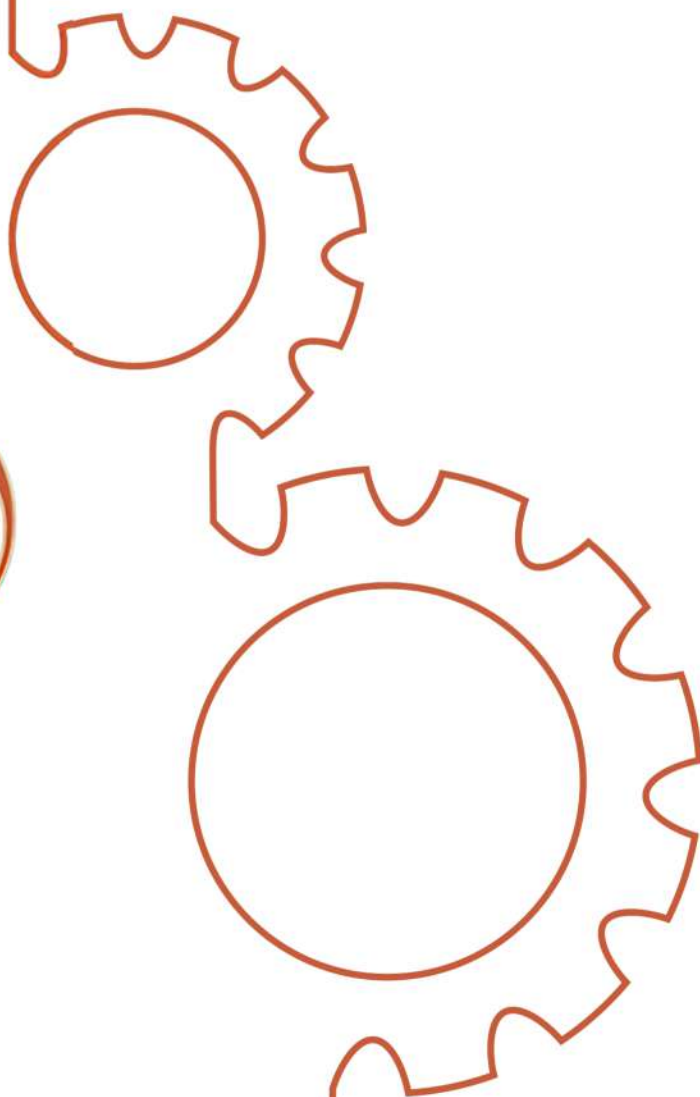


CHANDUBHAI S. PATEL INSTITUTE OF TECHNOLOGY

QUARTERLY NEWSLETTER

ENGINEERING

Insights



Jul 2025 - Sep 2025

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Message from Dean's Desk

I am pleased to share some of the exciting developments at the Chandubhai S. Patel Institute of Technology (CSPIT) in the recent past in our newsletter "**ENGINEERING INSIGHTS**".

The newsletter will focus on the inside activities, viz. academics, achievement of students and staff members as well as innovation occurring in the departments. In the era of engineering and technology, this newsletter will motivate the staff members and students on sharing their creativity and new ideas with the world and will help in their overall development.

As the saying goes by "**Coming together is a beginning, keeping together is progress and working together is success**". We hope that the readers find the issue informative and useful. I once again congratulate everyone for this issue of newsletter and wish the CSPIT a grand success.

Dr. Vijay Chaudhary
Dean, FTE



Message from Principal's Desk

I am delighted to present the fourth issue of “ENGINEERING INSIGHTS”, the quarterly newsletter of Chandubhai S. Patel Institute of Technology (CSPIT). This edition continues to showcase the vibrant activities, innovative initiatives, and remarkable achievements of our students and faculty in recent months. With a supportive management, experienced, dedicated and committed teaching and non-teaching staff, excellent infrastructure and upgraded facilities, the institution has achieved outstanding success since its inception in year 2000. The institute now strives and aspires to be a Centre of Excellence.

We are grateful to our Honourable Provost, Respected Registrar, distinguished Management members, Dean and Head of Departments for their constant support and encouragement. I express my sincere and heartfelt thanks to all the stakeholders staff members, students and parents who have co-operated and contributed towards the realization of our vision, mission and dreams. Let us march wholeheartedly in the journey towards actualization of our objectives and goals.

I would like to launch this issue with a quote **“The obstacles of the past can become the gateways to new beginnings, even a loss provides the polished direction of winning”**.

Dr. Trushit Upadhyaya
Principal, CSPIT

SPOTLIGHT ON MILESTONES

Kickstart @ CHARUSAT: Commencement Celebration 2025



The Faculty of Technology and Engineering (FTE), CHARUSAT organized the Commencement Celebration Programme for B.Tech first-year students of CSPIT and DEPSTAR during July 1–9, 2025. Over 700 students from CE, IT, CSE, AI & ML, EC, ME, EE, and CL branches participated in the orientation. With 30+ sessions by 20+ expert speakers, the program covered career opportunities, academic excellence, human values, spirituality, meditation, yoga, and creative activities like mind mapping, debate, and arts. Students also visited the “Amul Chocolate Plant, Mogar” and “Sardar Patel Memorial, Karamsad” gaining real-world exposure. The initiative received excellent feedback, fostering peer bonding, faculty interaction, and laying a strong foundation for students’ academic journey at CHARUSAT.



Satyanarayan Katha & Teachers' Day: A Day of Blessings & Gratitude

वक्रतुण्ड महाकाय सूर्यकोटि समप्रभः
निर्विघ्नं कुरु मे देव सर्वकार्येषु सर्वदा



श्रीगणेशाय नमः

Recharge & Reconnect: Staff Recreational Gathering | Legends of the Square (August 02, 2025)



Sports Fiesta (August 14, 2025)



Champions' Gallery: Sports Fiesta Faculty Winners



SPORT DAY
Faculty Celebration



EVENTS AND ACHIEVEMENTS

Expert Talk on “Semiconductor Opportunities, Applications, and Projects Future for Engineers”



The V. T. Patel Department of Electronics & Communication Engineering, in association with the IEEE CHARUSAT Student Branch, proudly inaugurated Gujarat's first IEEE CASS (Circuits and Systems Society) Student Branch Chapter on July 11, 2025. The ceremony marked a milestone in fostering innovation and research in circuit and system design, aligned with global semiconductor advancements. It featured an expert talk by Mr. Chintan Panchal, DFT Engineer, Einfochips, Ahmedabad on “Semiconductor Opportunities, Applications, and Projects Future for Engineers.” He highlighted the semiconductor industry's crucial role in shaping the digital future and the vast opportunities it offers aspiring engineers. Key discussions included VLSI, chip design, fabrication, and emerging areas like SoC (System-on-Chip) design, startups. The inauguration encouraged students and faculty to engage in IEEE CASS activities, drive interdisciplinary innovation, and bridge academia with industry to empower the next generation in circuits and systems.

Expert Talk on “Damp Proofing Methods and Materials”



M. S. Patel Department of Civil Engineering organized an expert talk on “Damp Proofing Methods and Materials” on July 11, 2025. The session was conducted by Mr. Sanjay Bhatt, Regional Manager Special Projects, Pidilite Industries Ltd., Ahmedabad, a seasoned expert with decades of experience in construction chemicals and waterproofing systems. The lecture was attended by 31 participants and aimed to bridge the gap between theoretical understanding and practical industry applications in the areas of damp prevention, structural protection, and durability enhancement. Mr. Bhatt discussed various chemicals and methods of waterproofing offered by Pidilite, explaining in detail the types of methods and materials to be used in different structural situations requiring waterproofing solutions, supported by real-world examples. The ease with which the speaker explained the topic made a lasting impression on the participants.

Expert Talk on "Advanced Technologies and Associated Challenges in Human Space Mission"



The CHAMOS Matrusanstha Department of Mechanical Engineering, together with the V. T. Patel Department of Electronics & Communication Engineering, hosted an expert talk, "Advanced Technologies and Associated Challenges in Human Space Mission" on August 4, 2025. The event was attended by both students and faculty members. The session featured Mr. Partho Ghosh, Scientist/Engineer-SD, SAC-ISRO, Ahmedabad, who provided an insightful overview of India's Gaganyaan Mission the country's maiden human spaceflight initiative, including the development of the Human-rated Launch Vehicle Mark-III (HLVM3), advanced Crew Escape System (CES), and detailed the crew module's design, thermal protection, environmental systems, and mechanical integration for astronaut safety. He also elaborated on the service module, highlighting its propulsion system, power management, and life-support mechanisms, along with key design considerations such as shock and vibration mitigation and system redundancy. The talk covered CES validation through TV-D1 abort trials and multi-stage parachute recovery tests. Mr. Ghosh also discussed upcoming uncrewed flights with Vyommitra humanoid, paving the way for India's vision of a permanent orbital facility under the Bhartiya Antrix Station initiative.

Expert Talk on "Job Opportunities and Current Trend in Silicon Industry"



The V. T. Patel Department of Electronics & Communication Engineering, organized an expert talk on "Job Opportunities and Current Trends in Silicon Industry" on August 5, 2025. The session was delivered by Mr. Nehal Jaiswal, Senior Software Engineer, AMD-Xilinx, Hyderabad. The expert talk explored strong job growth across the entire value chain, from design to manufacturing and testing, driven by emerging trends like AI, 5G, and IoT. The session also provided insights into the current era of Silicon Valley and its activities, equipping Electronics & Communication engineers to thrive in the rapidly evolving connectivity landscape.

Expert Talk on “Unlocking Monotonic Magic : Stack Based Problem Solving”



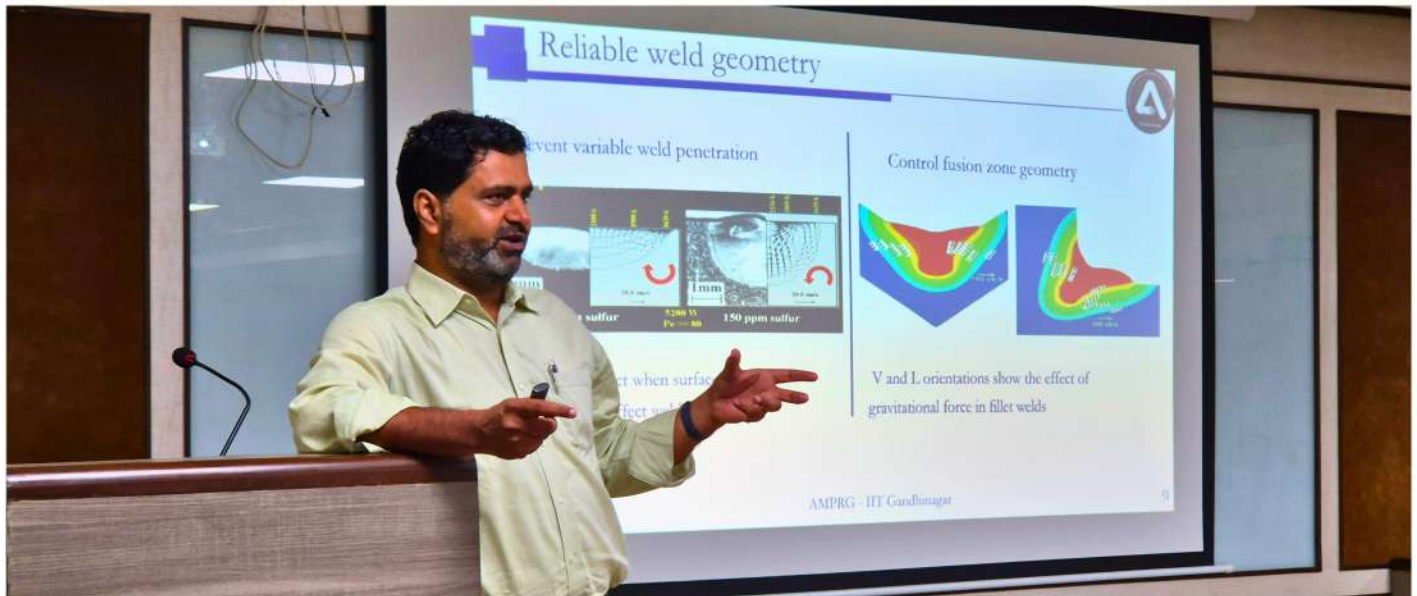
The Department of Computer Science & Engineering, organized an expert talk under the Mastering DSA & Cracking Interviews series, titled “Unlocking Monotonic Magic: Stack-Based Problem Solving” on August 6, 2025. The session was delivered by Ms. Isha Shah, Software Engineer at Microsoft, Hyderabad, who shared her rich industry experience and deep knowledge of Data Structures & Algorithms (DSA) with the students. Ms. Shah explained the concept of monotonic stacks, an advanced technique widely used to solve complex algorithmic challenges in coding competitions and technical interviews, and illustrated applications in popular problems such as Next Greater Element, Stock Span, and Sliding Window, highlighting their practical importance.

Expert Talk on “Modern Practices in DevOps Tools”



The K. D. Patel Department of Information Technology, organized an expert talk on “Modern Practices in DevOps Tools” on August 25, 2025. The session was arranged for B.Tech fifth Semester students and delivered by Dr. Harshad Prajapati, Associate Professor, DDIT, Nadiad. Dr. Harshad shared valuable insights on the evolving role of DevOps in the software development lifecycle, emphasizing automation, continuous integration, and deployment pipelines. He also highlighted industry trends, best practices, and the importance of adopting a DevOps culture for faster and more reliable software delivery. Students gained practical knowledge of real-world DevOps applications, bridging the gap between academic learning and industry requirements. The session witnessed active participation from students and faculty, with students appreciating the opportunity to interact with the expert and clarify their queries, while faculty encouraged them to implement the shared practices in their academic and project work.

Expert Talk on “Numerical Modeling of Welding and Additive Manufacturing”



CHAMOS Matrusanstha Department of Mechanical Engineering, organized an expert talk on “Numerical Modeling of Welding and Additive Manufacturing” on August 25, 2025, for second, third, and final year B.Tech. Mechanical Engineering students. The session was delivered by Dr. Amit Arora, Associate Professor in the Materials Engineering Department at IIT Gandhinagar, who also leads the Advanced Materials Processing Research Group. During the session, Dr. Arora provided an in-depth overview of the fundamentals and applications of numerical modeling in understanding welding and additive manufacturing. Through case studies and examples from his research group, he showcased how numerical simulations are used to design new alloys, optimize printing strategies, and achieve high-performance components. The talk concluded with a vote of thanks, acknowledging Dr. Arora’s efforts in sharing his expertise and motivating students to explore research in welding, additive manufacturing, and simulation-based engineering.

Expert Talk on “Igniting Chip Careers: From Research Labs to Semiconductor Startups”



The V. T. Patel Department of Electronics & Communication Engineering, in collaboration with the IEEE Circuits and Systems Society (CASS) Student Branch Chapter – CHARUSAT and ChipXplorer – VLSI Club, organized an expert talk titled “Igniting Chip Careers: From Research Labs to Semiconductor Startups” on August 25, 2025. The session, part of the CASS Convergence 2025 series, was delivered by Mr. Marmik Bhatt, Founder of Monk9 Technologies, Rajkot. Attended by 121 students, the talk provided in-depth insights into the journey from research and innovation in laboratories to successfully launching semiconductor-based startups. Mr. Bhatt shared practical guidance on identifying opportunities, navigating challenges, and positioning oneself in the competitive global semiconductor ecosystem, motivating students to explore innovation-driven entrepreneurship beyond traditional career paths and contribute to India’s growing startup ecosystem.

Expert Talk on "Device Physics Meets Design: Understanding the Heart of VLSI"



The V. T. Patel Department of Electronics & Communication Engineering, in collaboration with the IEEE Circuits and Systems Society (CASS) Student Branch Chapter – CHARUSAT and ChipXplorer – VLSI Club, hosted an expert talk titled "Device Physics Meets Design: Understanding the Heart of VLSI" on August 30, 2025, as a part of the CASS Convergence 2025 initiative. The session was delivered by Prof. Harpreet Singh Jattana, Professor of Practice at DA-IICT, Gandhinagar and Group Head at the Semi-Conductor Laboratory (SCL), Mohali. Attended by 118 students, the talk highlighted how device physics forms the foundation of efficient VLSI design and provided practical insights into the physics shaping next-generation semiconductor technologies.

Expert Talk on "Opportunities as an Entrepreneur for Engineers"



The M. S. Patel Department of Civil Engineering, under the aegis of the Indian Concrete Institute (ICI) Student Chapter, CHARUSAT, organized an expert talk on "Opportunities as an Entrepreneur for Engineers" on September 1, 2025. The session was conducted by Mr. Haituk Patel, Founder and Chief Strat-x-preneur of Management Designers, Vadodara, a boutique business consulting firm, and a distinguished entrepreneur and business strategist known for his innovative approaches to business growth and management. During the session, Mr. Haituk provided valuable insights into the importance of entrepreneurship for engineering graduates, key skills and competencies needed to start and sustain a business, emerging opportunities and challenges in the current industry landscape, practical steps to transition from students to successful entrepreneurs, and real-life examples and case studies from his own experiences. The talk encouraged students to develop an entrepreneurial mindset, emphasizing adaptability, continuous learning, and networking, and included a Q&A session where Mr. Haituk clarified doubts regarding startup processes, funding, and scaling.

Expert Talk on "Artificial Intelligence and Machine Learning"



The Department of Artificial Intelligence and Machine Learning organized an expert talk on "Artificial Intelligence and Machine Learning" on September 9, 2025. The session was conducted by Mr. Prasanjit Das, SAMATRIX, Gurgaon, and was specially designed to provide a strong foundation in these emerging technologies and their real-world impact. Delivered in an interactive format, the session introduced the fundamentals of Artificial Intelligence, explaining how machines simulate human intelligence through learning and decision-making. It further explored the concept of Machine Learning, focusing on how algorithms improve performance by learning from data. The speaker also emphasized the importance of these technologies in driving automation, efficiency, and improved decision-making, along with career opportunities in roles such as Data Scientist, Machine Learning Engineer, and AI Researcher. The event proved highly engaging, equipping students with conceptual clarity, future career directions, and inspiration to explore the transformative potential of AI & ML.

Expert Talk on "Understanding of Chip Level Designing for UAVs"



The V. T. Patel Department of Electronics & Communication Engineering, organized an expert talk on "Understanding of Chip Level Designing for UAVs" under the Drone Club – Adwitiya Aakash on July 10, 2025. The session was delivered by Dr. Kaushal Vala, Senior Engineer at Cadence, Ahmedabad. The talk focused on chip-level designing of UAVs (Unmanned Aerial Vehicles), which refers to the process of creating and integrating electronic components directly at the semiconductor level. Dr. Vala explained the importance of custom microcontrollers, SoC (System-on-Chip), FPGA (Field Programmable Gate Array), and ASIC (Application Specific Integrated Circuit) in enhancing the performance, efficiency, and functionality of UAVs, providing students with a deep understanding of the technologies and design approaches used in advanced aerospace applications.

Workshop on “.NET Framework and C#”



The U & P U. Patel Department of Computer Engineering, organized a workshop on “.NET Framework and C#” on July 5, 2025. More than 50 students from the Artificial Intelligence and Machine Learning, Computer Engineering, and Information Technology branches took part in the workshop. The session was delivered by Mr. Gagan Dubey, Corporate Trainer at Rishabh Software, Vadodara. He provided practical demonstrations of ASP .NET web application development and C# object-oriented concepts using real-world industry cases and examples. The workshop offered students hands-on exposure to the application of modern programming frameworks and coding practices, helping them connect academic learning with industry requirements.

Workshop on “Code to Cloud: Hands-on with Embedded IoT & Visual Programming”



The K. D. Patel Department of Information Technology, organized a two-day workshop on “Code to Cloud: Hands-on with Embedded IoT & Visual Programming” on July 19, 2025, and August 2, 2025. The sessions were conducted by Dr. Jignesh Patoliya, Senior Engineer at eInfochips Ltd., Ahmedabad, who shared insights into IoT-based solution development. On the first day, participants learned the fundamentals of the Internet of Things (IoT), embedded system design, and visual programming tools, covering IoT architecture, sensor integration, and coding practices. On the second day, participants engaged in cloud integration for IoT devices through hands-on demonstrations, connecting embedded systems with cloud platforms, managing data transmission, and implementing real-time monitoring. A total of 50 students participated enthusiastically, gaining practical skills, innovative thinking, and industry-oriented knowledge to support their future careers.

Hands-on Session on “HTML & Bootstrap”



The Department of Artificial Intelligence and Machine Learning, conducted a hands-on workshop titled “HTML & Bootstrap” on July 24, 2025 for B.Tech third semester students to bridge the gap between basic web development and responsive design. The session was delivered by Mr. Parag Patel, Shri Om Infotech, Ahmedabad, who shared industry insights and real-world applications of Bootstrap in web development. The session enhanced students’ knowledge of HTML and guided them in using Bootstrap, a popular front-end framework. Students learned about Bootstrap components, the responsive grid system, utility classes, and how to convert static HTML pages into responsive websites. The workshop included live demonstrations, coding sessions, and interactive discussions for better understanding. A total of 68 participants attended, gaining practical skills to support their academic and professional projects.

Workshop on “DevOps in Practice: From Agile Pipelines to Secure Deployments”



The Department of Computer Science & Engineering, organized a workshop on “DevOps in Practice: From Agile Pipelines to Secure Deployments” on August 20, 2025. Conducted under the banner of Cyber Kavach: Cyber Security Club in offline mode, the session was led by Mr. Binank Patel, Technical Lead at Electromech Cloud Pvt. Ltd., Ahmedabad. The workshop provided participants with a comprehensive understanding of modern DevOps practices, focusing on integrating Agile pipelines, implementing CI/CD workflows, and ensuring secure deployments in contemporary IT environments. The session emphasized the critical role of DevSecOps in embedding security at every stage of the software development lifecycle. The event witnessed enthusiastic participation from 110 students, who actively engaged in learning and applying concepts, with feedback reflecting high appreciation for industry-relevant insights, interactive sessions, and practical exposure.

Workshop on “Data Science with Azure AI”



The V. T. Patel Department of Electronics & Communication Engineering, organized a two-day workshop on “Data Science with Azure AI” on August 20–21, 2025. The workshop was led by Mr. Jigar Pathak, Cloud Solution Architect and Microsoft Certified Trainer, Partner at TechScalable, Pune, and engaged 125 students in exploring data fundamentals, cloud-based analytics, and AI applications. The sessions covered concepts of machine learning applied through Microsoft Azure, as well as tools such as Microsoft Fabric, Azure Databricks, Power BI, and Cognitive Services, along with hands-on labs on deploying ML models and building chatbots. The workshop provided students with practical exposure to modern data science and AI tools, equipping them with skills for real-world problem-solving.

Workshop on “Linux Kernel Compilation”



The U & P U. Patel Department of Computer Engineering and K. D. Patel Department of Information Technology jointly organized a “Linux Kernel Compilation” workshop on September 4, 2025. The session was conducted by Dr. Hemal Shah, Project Manager at eInfochips, Ahmedabad, who guided students through the concepts and practical aspects of kernel compilation. A total of 33 students actively participated in the workshop. The workshop provides hands-on exposure to the fundamentals of building and customizing the Linux kernel. The session covered key topics, including the role of Make files, creation of static and dynamic libraries, kernel architecture, and the process of kernel compilation and cross-compilation. Participants gained practical insights into how the Linux kernel is developed and tailored for different platforms, preparing them for real-world applications in systems programming, embedded development, and advanced computing environments.

FDP on “Bridging Academia and Industry: Emerging Trends and Practical Insights in IoT”



The V. T. Patel Department of Electronics & Communication Engineering, organized a Faculty Development Program (FDP) on “Bridging Academia and Industry: Emerging Trends and Practical Insights in IoT” during June 20–21, 2025. The sessions were conducted by Mr. Hardik Patel, CEO of Epsilon Electronics, Ahmedabad, and Mr. Dhruv Shah, Senior Firmware Developer at Volansys – ACL Digital, Ahmedabad. Mr. Hardik Patel covered the recent trends in IoT in the industry, including a hands-on session on the MQTT protocol, while Mr. Dhruv Shah discussed the Matter protocol, highlighting new pathways for IoT-based home automation designs. Participants actively explored the latest IoT technologies through practical, hands-on sessions, gaining valuable insights and experience to bridge academic learning with industry applications.

THM X CHARUSAT Chapter & THM-VADODARA Cybersecurity Meetup



The Department of Computer Science & Engineering, proudly hosted the “THM X CHARUSAT Chapter & THM-VADODARA Cybersecurity Meetup” on July 18, 2025. The event featured distinguished speakers Mr. Harshil Parikh, Mr. Chirag Shah, Mr. Prabhakar Damor, and Mr. Pragnesh Singh Rajput from the THM Vadodara Chapter, who delivered insightful sessions on key cybersecurity topics, including the hacker’s mindset, personal security, and digital hygiene. Over 100 participants attended the event, gaining industry-relevant knowledge and practical skills. The meetup concluded with an engaging debate and a hands-on Capture the Flag (CTF) challenge. Participants gained valuable insights into the global cybersecurity market, current threat scenarios, and the impact of emerging technologies, such as Artificial Intelligence, on system security. The event successfully achieved its objective of promoting awareness and skill development in cybersecurity, serving as a meaningful platform for knowledge exchange, industry connection, and career-oriented guidance.

Workshop on “Master Modern JavaScript Frameworks”



The Department of Computer Science & Engineering, organized a workshop titled “Master Modern JavaScript Frameworks” on August 2, 2025, to enhance students’ understanding of frontend development and align academic learning with industry practices. The session was conducted by Mr. Parth Mehta, Senior Software Developer at Sapphire Software Solutions, Ahmedabad, in collaboration with the TechGenius Student Club and the IEEE Student Branch. The workshop provided hands-on exposure to modern JavaScript frameworks and tools, including jQuery for simplified DOM manipulation and event handling. Through live demonstrations, coding exercises, and practical examples, participants strengthened their knowledge of building scalable, and interactive web applications. A total of 38 students actively participated, valuing the industry insights and interactive teaching that kept them engaged throughout the session.

Hackathon : “Tech Tackle”



The V. T. Patel Department of Electronics & Communication Engineering, through its Code for Cause club, proudly organized a hackathon titled “Tech Tackle” on August 30, 2025, where innovation met collaboration. A total of 93 students from the Electronics & Communication, Computer Engineering, Information Technology, Computer Science & Engineering, and Artificial Intelligence and Machine Learning departments participated, forming teams to brainstorm, build, and showcase exciting IoT-based projects using ESP32 and Arduino Uno boards. From ideation to execution, participants demonstrated exceptional problem-solving skills, creativity, and teamwork. The hackathon provided a platform to apply technical knowledge to real-world IoT solutions, gain hands-on experience with Arduino and ESP32 hardware, enhance teamwork, communication, and leadership skills, strengthen problem-solving and innovative thinking, and understand the complete journey from idea to prototype, effectively bridging the gap between classroom learning and practical applications.

Alumni Talks – Insights from Our Esteemed Graduates

Aatrey Vyas, Senior GPU Software Developer Engineer, Intel

Mr. Aatrey Vyas, GPU Software Development Engineer at Intel, Bangalore, and alumnus of CE-CSPIT, delivered an insightful industry talk on “Unlocking Career Opportunities: A Guide for Aspiring Professionals” on July 18, 2025. He shared his inspiring journey from student life to a successful career in the tech industry, along with practical tips on resume building, placement preparation, and navigating global job markets.



Mr. Yash Patel, Security Engineer, Microsoft

Mr. Yash Patel, working at Microsoft, USA and alumnus of CE-CSPIT delivered an expert session on “Secure-By-Design: Threat Modeling & Resilience in Modern IT Systems” on August 2, 2025. He shared his deep knowledge and practical insights on building secure and resilient IT infrastructures, emphasizing threat modeling techniques and security practices. The session provided students with a clear understanding of current industry challenges and strategies to address them.



Mr. Darshan Kansagara, Software Engineer, Apple Inc.

Mr. Darshan Kansagara, Software Engineer at Apple Inc., Hyderabad, and alumnus of CE-CSPIT conducted a career guidance session on July 9, 2025. He shared his professional experiences and offered valuable advice on career planning, skill development, and navigating opportunities in the tech industry. His insights helped students better understand industry expectations and prepare effectively for their future roles. The session was engaging, interactive, and highly beneficial for students aspiring to build successful careers.



Mr. Anmol Chauhan, Data Scientist, Odysseus Solutions

Mr. Anmol Chauhan, Data Scientist at Odysseus Solutions, USA, and alumnus of CSE-CSPIT, delivered a career guidance session for B.Tech fifth semester students on August 22, 2025. His real-world experiences helped students understand industry expectations and prepare for future challenges. The interactive session, attended by 53 students, provided valuable insights and motivated participants to enhance their career readiness.



Mr. Jugal Doshi, BIM Engineer, Systra India Pvt. Ltd.

Mr. Jugal Doshi, BIM Engineer at Systra India Pvt. Ltd., Ahmedabad and alumnus of Civil-CSPIT, conducted a workshop on “Revit Family Creation” on August 2, 2025, organized under the ICI and IAstructE Student Chapters. The session provided hands-on guidance on preparing project documents and Bills of Quantities for various construction activities. He also emphasized the significance of maintaining detailed checklists for effective quality control in construction projects.



Alumni Talks – Insights from Our Esteemed Graduates

Mr. Sagar Joshi, Deputy Manager, Polycab India Ltd.

Mr. Sagar Joshi, Deputy Manager at Polycab India Ltd., Vadodara, and alumnus of EE-CSPIT, delivered an expert talk on “Inside the Cable: Materials, Design Principles, and Construction Techniques” on July 23, 2025. The session provided students with an in-depth understanding of cable materials, insulation, shielding, and design factors. Practical demonstrations on construction techniques like stranding and armouring helped bridge theory with industry practices.



Mr. Ayush Ghadiya, AI Research Engineer, Sony

Mr. Ayush Ghadiya, AI Research Engineer at Sony and alumnus of IT-CSPIT, delivered an Alumni Expert Talk on “Shaping Your Future with AI” on August 23, 2025. Organized by the K. D. Patel Department of Information Technology, under the Data Science Club, the session highlighted emerging trends and career opportunities in Artificial Intelligence. With 40+ enthusiastic participants, the event inspired students to explore AI-driven careers and provided valuable guidance for shaping their professional journey.



Ms. Palak Jadhav, Director – UX, Lemon Yellow LLP

Ms. Palak Jadhav, Director – UX at Lemon Yellow LLP, Thane, Maharashtra, and alumnus of IT-CSPIT, conducted a workshop on “Hands-on Design Thinking & User Experience (UX): Bridging Empathy with Logic” on September 16, 2025. Organized by the K. D. Patel Department of Information Technology, the session introduced 47 participants to modern UI/UX principles and key design laws such as Hick’s Law, Miller’s Law, and Fitts’s Law. Students also gained practical exposure to Figma, combining theory with hands-on learning to strengthen their understanding of user-centered design.



Mr. Hardik Patel, Founder & CEO, Epsilon

Mr. Hardik Patel, Founder & CEO of Epsilon and alumnus of EC-CSPIT, delivered an inspiring Alumni Expert Talk on “The Skill Revolution: Education for Innovation” on August 30, 2025. Organized by the V. T. Patel Department of Electronics & Communication Engineering, the session engaged 90 students and focused on emerging job opportunities, the impact of AI and quantum computing, essential entrepreneurial skills, and start-up challenges. The interactive session concluded with a lively Q&A, where students received valuable guidance and practical insights from the expert.



Ms. Vrunda Patel, AI Engineer-Data Scientist, IBM EELab

Ms. Vrunda Patel, AI Engineer – Data Scientist at IBM EELab and an alumnus of IT-CSPIT, conducted a workshop on “Hands-on Roadmap to Agentic AI” on August 04–05, 2025, organized by the Artificial Intelligence and Machine Learning Department. The session introduced 56 participants to intelligent agents, autonomous reasoning, and real-world applications, with guided activities offering practical exposure to agent-based systems and the future of AI. Through guided activities and expert insights, students gained practical exposure to agent-based systems and strengthened their understanding of how Agentic AI is shaping the future of artificial intelligence across industries.



Student Spotlight: Rising Stars

ISRO Robotics Challenge – IRoC-U 2025



The Chandubhai S. Patel Institute of Technology, CHARUSAT, proudly announces that its students represented the institute as Team AVIONIX in the ISRO Robotics Challenge – IRoC-U 2025. The team, comprising of Dharmil Patel (22ME013), Siddhi Pandya (22IT084), Vyom Pandya (22IT157), and Aum Barai (22AIML002), was among the top 16 finalists selected from over 500 teams nationwide and the only team from Gujarat. Organized by the U R Rao Satellite Centre (URSC), ISRO, the competition encourages youth to design UAVs for planetary exploration. Team AVIONIX presented their prototype drone before ISRO experts at URSC, Bengaluru, and later showcased their work at National Meet 2.0 and National Space Day 2025 in New Delhi, where they interacted with scientists and space leaders, including Group Captain Angad Pratap of India's first manned Moon mission. This remarkable achievement highlights CHARUSAT's commitment to fostering innovation and excellence in space science and technology.



Dewang Mehta IT Awards

Rohan Vaghela (22CS088), Tanvi Mehta (23CS044), Diya Gangurde (23EC031), Dev Shah (23IT116), Bhumi Patel (22IT091), Hiren Modhvadia (22AIML023), Hari Patel (23AIML049), Hitarth Khatiwala (24AIML019) students from CSPIT were honored at the Prestigious Dewang Mehta IT Awards, showcasing their innovation and technical excellence.

Student Spotlight: Rising Stars

Team Ojaswat in DKDC - 2025



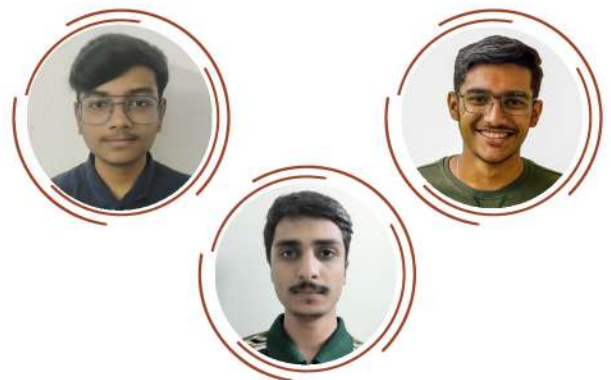
University Formula Team Ojaswat Motorsports participated and performed fantastically at Dirt Kart Design Challenge (DKDC) - 2025 organised by Fraternity of Mechanical and Automobile Engineering (FMAE) during July 1 - 3, 2025 at Dirt Arena track, Valpoy, Goa. Team secured Winner position in Design Presentation, Runners up position in Acceleration, Runners up position in Endurance and Fuel Economy, Overall Runners up at DKDC IC (Internal Combustion) Engine category. Engineering Enthusiastic Students from Mechanical, Electrical and Electronics Communication streams had designed and manufactured the Kart in-house in a tenure of one year after learning software's such as

Ansys, Solidworks, MATLAB, KiCAD, etc. More than 22 teams were registered for the same event out of which only 9 teams reported for Technical Inspection. Only 3 teams cleared all necessary tests and TI to run for Endurance Race. Team students learnt to collaborate, communicate, procure and establish contacts with industrialists. Team Ojaswat Motorsports was sponsored by Wadia, Alpha cube, Manish EngiCorp., Nexus Engitech, Solidworks and Ansys. Team performed well in the terms of Technical, as well as other Non-Technical aspects like Person in Demand for the Team, Team Selection, Department Allocation, Material Procurement, Team Management, Inventory, Logistics, Financial Management and Many More.

Winning is just part of team's journey, but the learning experience is crucial which students will take from this team throughout their life!

NEXOTHON

A team from the Computer Engineering Department, consisting of Nemin Haria (23CE040), Jay Dolar (D24CE187) and Bhavya Sonigra (D24CE188), earned the 2nd Runner-Up title at NEXOTHON hosted by GCET, CVM University, Vallabh Vidyanagar. The team was rewarded ₹5000 cash prize.



Tech-Tonic '25

The team from Computer Science and Engineering Department, comprising of Dhairya Kanabar (23CS032), Tanvi Mehta (23CS044), Mitul Mistry (23CS045), and Kathan Modh (23CS046), clinched 1st Rank at the Tech-Tonic Hackathon 2025 with their innovative solutions.

Student Spotlight: Rising Stars

Tech-Tackle '25

The team from the Department of Information Technology, consisting of Abhi Kakadiya (24IT035), Priyaben Kheni (24IT041), and Heer Ghevariya (24IT025), secured the 2nd Rank in Tech tackle 2025, demonstrating their problem-solving and technical skills in the hackathon.



The Computer Science and Engineering Department team, consisting of Pankti Akbari (23CS001), Yash Bhavada (23CS006), and Shashan Lumbhani (23CS042), secured the 3rd Rank in the Tectackle event. They were rewarded with a cash prize of ₹3000 for their problem-solving skills.

Oracle Certification

Rohit Harshkumar (22IT132) and Prajapati Prem (22IT127) from the Information Technology Department achieved the Oracle Certified Professional – AI Vector Search (2025) certification, demonstrating their advanced knowledge in emerging AI technologies.



Celebrating Excellence: Faculty Achievements

Network Defender Certified | SQLite Forensics with Belkasoft | Top 4% in HACKERVERSE



Dr. Priteshkumar Prajapati from the Information Technology Department achieved the Certified Network Defender credential, SQLite Forensics with Belkasoft and secured a top 4% rank in the Hackverse CTF Challenge, showcasing his strong cybersecurity expertise.

AWS Certified AI Practitioner

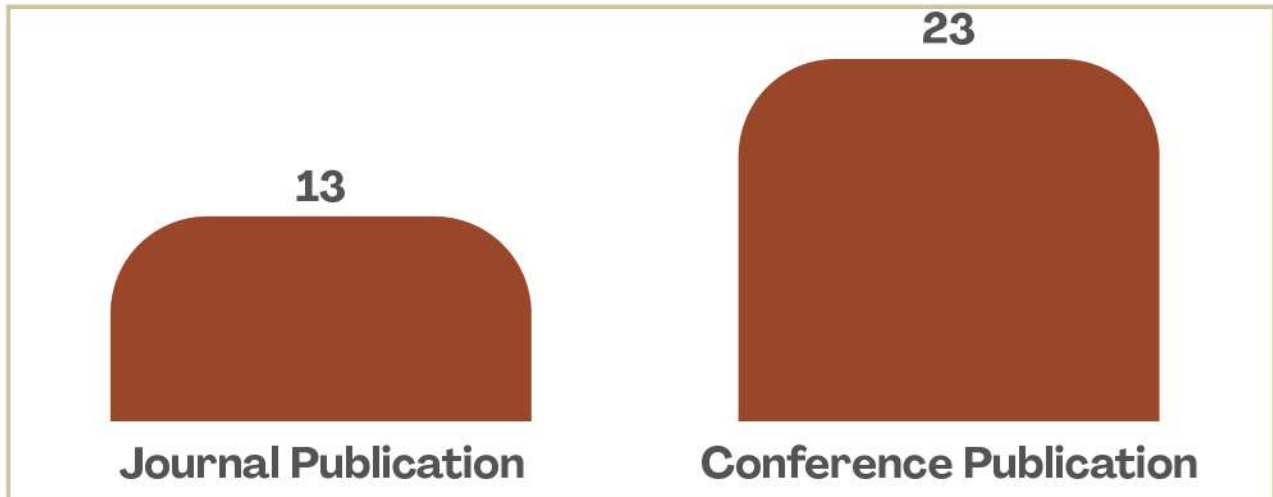
Dr. Sanket Suthar from the Information Technology Department earned the AWS Certified AI Practitioner credential, demonstrating his expertise in applying artificial intelligence solutions using AWS technologies.



Best Paper Presentation Award



Prof. Vikas Panchal from the Mechanical Engineering Department received the Best Paper Presentation Award under the Faculty Category at the Second International Conference on Management, Science & Technology (ICMSTII) – 2025 for his research paper titled “Nano-Coatings on Piston Rings: Advances, Tribological Performance, and Impact on Internal Combustion Engine Efficiency”.

RESEARCH AND INNOVATION**Academic Publications Analysis: Indexed Achievements in Research**

For the academic year 2024–25, CSPIT recorded an impressive total of 79 journal publications and 213 conference publications, reflecting the institute's strong research culture and commitment to scholarly excellence. Continuing this momentum, CSPIT further strengthened its research endeavors with remarkable achievements during the quarter of July to September 2025. A total of 13 research papers were published in reputed journals indexed in Scopus and Web of Science, demonstrating the faculty members' dedication to quality and impactful research, while 23 papers were presented and published in international conferences, providing global recognition and visibility to CSPIT's research community. These accomplishments highlight the institute's continuous pursuit of academic excellence, innovation, and contribution to the advancement of science and technology at both national and international levels.

Paper Publication in High Impact Factor Journals (Selected)



Dr. Bhavin Mehta published a paper titled "Performance and Emission Evaluation of Diesel Engines Using Karanja Biodiesel Blended with Diethyl Carbonate" in the Journal "Results in Engineering" having JCR Impact factor of 7.9.

Dr. Jigar Sarda published a paper titled "Optimizing object detection for autonomous robots: a comparative analysis of YOLO models" in the Elsevier based Journal "Measurement" having JCR Impact factor of 5.6.



Dr. Kawaljitsingh Randhawa published a paper titled "Organic membranes in current separation technologies: Materials, fabrication, and economic perspectives" in the Journal "Inorganic Chemistry Communications" having JCR Impact factor of 5.4.

Dr. Mayur Sutaria published a paper titled "Evaluating the wear and friction characteristics of heat-treated and as-cast ex-situ Al-Si7Mg/SiC composites under lubricated conditions" in the journal "Discover Materials" having JCR Impact factor of 5.1.



Dr. Punit Patel published a paper titled "Microstructural damage analysis during drilling in composite laminate with and without smart magneto-elastomer backup" in the Journal "Scientific Reports" having JCR Impact factor of 3.9.

Prof. Mayur Makwana published a paper titled "Enhanced WDM-PON architecture with SOA, EDFA, and Raman amplifiers for next-generation optical networks" in the Journal "Optics" with a JCR Impact Factor of 2.5.



Dr. Nilay Patel published a paper titled "Enhanced Voltage Stability and Fault Ride-Through Capability in Wind Energy Systems using FACTS Device Integration" in the MDPI based Journal "Machines" having JCR Impact factor of 2.5.

CARRIER & PLACEMENTS



Visiting Companies Jul 25 - Sep 25



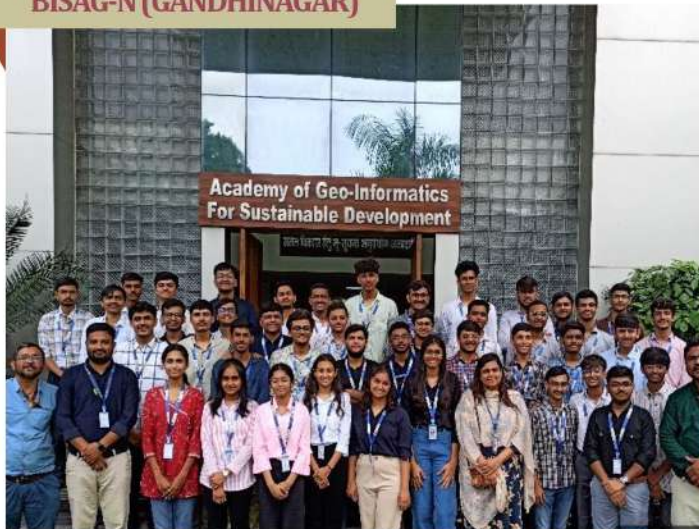
INDUSTRY COLLABORATIONS

Student Industrial Tours and Visits

As an essential part of engineering education, CSPIT organized industrial visits for students of all disciplines, as they provide practical exposure to real-world industrial processes, manufacturing techniques, and professional work environments.



BISAG-N (GANDHINAGAR)



The Department of Computer Science & Engineering and the U & P U. Patel Department of Computer Engineering, organized an industrial visit to Bisag-N, Gandhinagar, on August 20-22 and 28-29, 2025. The visit was designed to provide practical exposure to B.Tech third semester students, enabling them to understand the real-world applications of their academic learning. A total of 229 students actively participated in the visit, gaining valuable industry insights and hands-on experience. The visit successfully bridged the gap between classroom knowledge and modern industry practices, significantly enhancing the students' technical and professional outlook.

During the visit, students gained valuable insights into cutting-edge technologies such as satellite communication, remote sensing, geo-informatics, and software development for space applications. Experts at BISAG provided demonstrations on real-time projects and applications, helping students to understand the practical implementation of concepts learned in the classroom. This visit significantly enhanced their technical knowledge and exposure to research-driven projects, fostering interest in advanced technologies and their role in national development.



Student Industrial Tours and Visits

GERI (VADODARA)



The M. S. Patel Department of Civil Engineering organized a technical visit to the Gujarat Engineering Research Institute (GERI), Gotri, Vadodara, on July 19, 2025 for B.Tech final year students. The visit was aimed to provide students with hands-on exposure to state-of-the-art testing methods in soil mechanics, construction materials, and structural components, along with a comprehensive understanding of the design and performance of water resource structures. GERI, a premier research and testing institute under the Government of Gujarat, plays a vital role in advancing hydraulic and geotechnical engineering practices.

An industrial visit to Kotarpur Water Treatment Plant (WTP), Ahmedabad, was organized on August 29, 2025, for second, third, and fourth year Civil Engineering B.Tech students. The visit was aimed to provide students with practical exposure to advanced and sustainable water treatment methods and the processes involved in supplying safe and potable water to urban households. Students learned about various treatment processes including pre-chlorination, alum dosing, clari-flocculation, filter bed treatment, post-chlorination, solid sludge removal, and storage in clear water reservoirs.

KOTARPUR WTP (AHMEDABAD)



SHETRUNJI DAM (PALITANA)



A technical field visit to the Shetrunji Dam, in the Bhavnagar district, was organized on July 25, 2025, for Civil Engineering students. The visit provided students with practical exposure to dam engineering and hydraulic systems. The dam, a masonry-cum-earthen structure for irrigation and water supply, features gravity spillways, energy dissipation systems, and controlled releases to manage floods and regulate flow.

On July 9, 2025, the M & V Patel Department of Electrical Engineering organized an industrial visit to Kloeckner DESMA Machinery Pvt. Ltd., Ahmedabad, for fifth and seventh semester B.Tech students. The visit provided practical exposure to industrial electrical systems, automation, and control technologies. Students toured the production plant, observing control panel design, motor control, PLC-based automation, and safety practices.

DESMA MACHINERY PVT. LTD (AHMEDABAD)



STUDENT SUPPORT AND DEVELOPMENT**Empowering Students Through Financial Assistance: UGSF/PGSF's Commitment to Excellence**

UGSF/PGSF

Student Fellowship

28 Students

Financial Support

Rs. 2,72,600

The Undergraduate & Postgraduate Student Fellowship (UGSF & PGSF) provides financial support to help students pursue their academic goals, research projects, and skill development. This assistance allows students to focus on their studies without financial worries. By fostering an inclusive environment, UGSF & PGSF promotes innovation, critical thinking, and professional growth, helping students reach their full potential and contribute to a better future.

University Financial Support for Students Conference Publication

CONFERENCE

Publication

7 Students

Financial Support

Rs. 94,910

The university supports students by covering expenses for paper publication and conference participation, including registration fees, travel, and accommodation. This assistance enables students to present their research on national and international platforms, promoting academic growth, knowledge exchange, and networking. By removing financial barriers, the university helps students share their work, gain exposure to advanced research, and build their academic and professional profiles, fostering their research careers and overall success.

EXTRA-CURRICULAR ACTIVITIES

NSS Activity: "Raksha Bandhan – Tribute to Our Brave Soldiers"



The V. T. Patel Department of Electronics & Communication Engineering, organized an event titled "Raksha Bandhan – Tribute to Our Brave Soldiers" on July 28, 2025. The event aimed to celebrate the spirit of Raksha Bandhan by expressing gratitude and emotional support to the soldiers of the Indian Armed Forces, who protect the nation away from their families. A total of 50 participants, including 39 volunteers and 7 faculty members, enthusiastically took part in making handmade rakhis and writing heartfelt letters filled with encouragement, prayers, and appreciation. The initiative instilled a sense of empathy, patriotism, and social responsibility among students while offering a meaningful way to honor the sacrifices of soldiers and strengthen the bond between the youth and the nation's protectors.

NSS Activity: "STEM Awareness"



On August 6, 2025, IEEE CHARUSAT Student Branch WIE Affinity Group, in collaboration with NSS CHARUSAT and organized by the Department of Computer Science & Engineering, conducted a STEM Awareness event at S. I. Patel Ipcowala Madhyamik Vidhyalaya, Roon Village. The event aimed to inspire rural school girls in Science, Technology, Engineering, and Mathematics through motivational talks and fun activities like the Tower of Hanoi puzzle, STEM crosswords, and an If-Else Loop game. Simple engineering projects were demonstrated, and the program concluded with a hand-printing session symbolizing the students' enthusiasm.

WDC activity: EmpowerHER – Unlocking Financial Freedom

The Women Development Cell (WDC) of CE, CSE, IT, and AIML departments of CSPIT jointly organized an event titled “EmpowerHER – Unlocking Financial Freedom” on August 26, 2025. The session featured esteemed speakers Ms. Rachana Oza, SEBI-Certified Securities Market Trainer and Sr. Project Manager at LakshMe, Ahmedabad and Ms. Heli Miyani, NISM-Certified Financial Literacy Expert and Community Manager at LakshMe, Ahmedabad. They shared valuable knowledge on financial literacy, investment awareness, and strategies for achieving financial independence. The event witnessed the enthusiastic participation of 200 girl students of CHARUSAT, making it a highly impactful and insightful session.

ENGINEERING *Insights*

VISION

To become a leading institute for creation and dissemination of knowledge in the frontiers of technology.

MISSION

To prepare world-class technocrats and researchers and facilitate enhanced deployment of technology for betterment of lives.



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